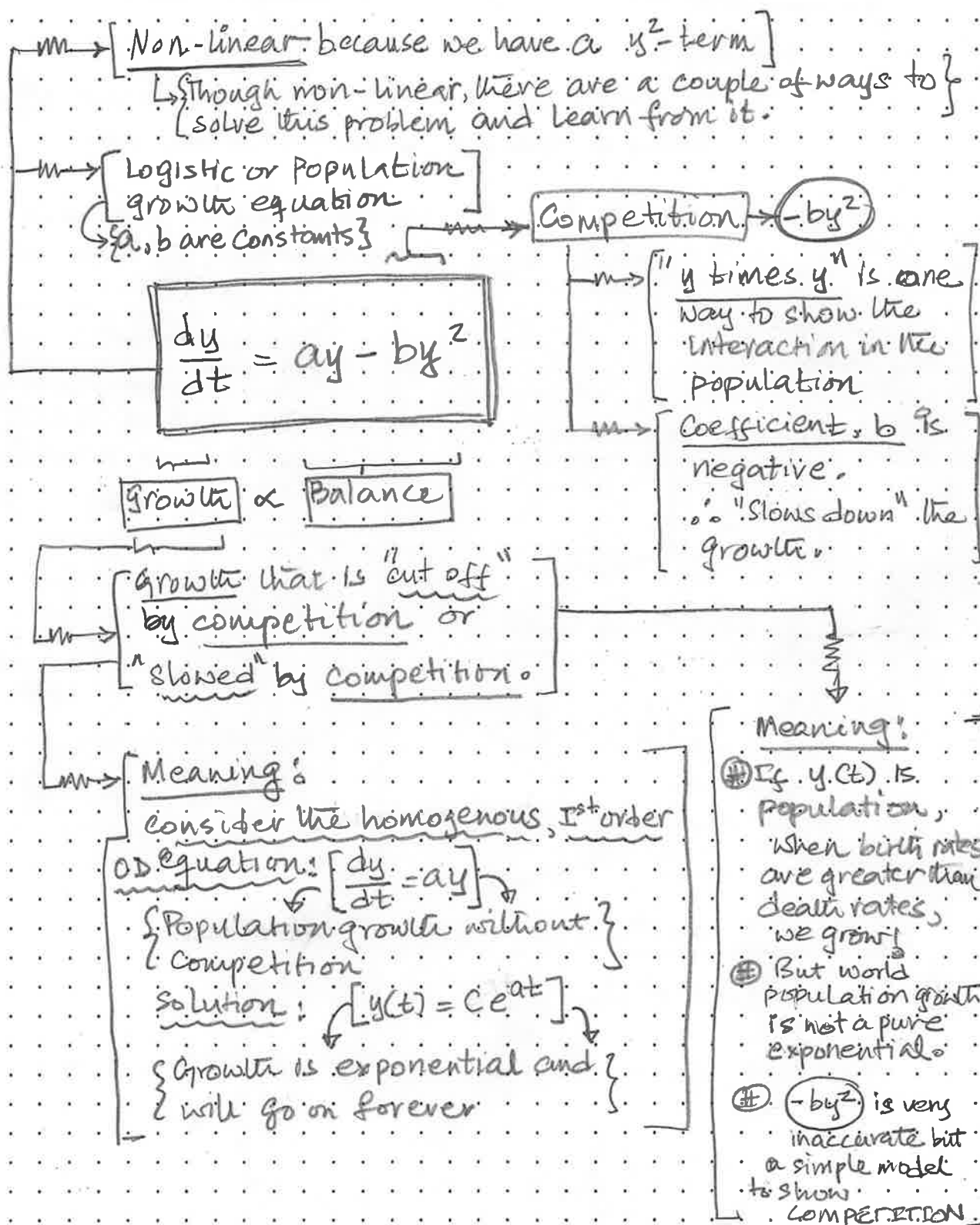
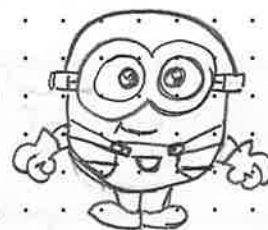


SIMPLEST FIRST-ORDER NON-LINEAR EQUATION

LOGISTIC / POPULATION GROWTH EQUATION



SOLUTION TO $\left[\frac{dy}{dt} = ay - by^2 \right]$



"Neat trick" to solve this non-linear equation.

GOALS

1. Do something to make this non-linear, 1st order ODE, a simple, linear, 1st order ODE.
2. We need a systematic way to work with this "special" type of non-linear, 1st order ODE.

Solution

① START BY LETTING,

$$z = \frac{1}{y}$$

② DIFFERENTIATE EQ ①:

$$\frac{dz}{dt} = \frac{d}{dt} \left(\frac{1}{y} \right) = \frac{d}{dt} (y^{-1})$$

$$\Rightarrow \frac{dz}{dt} = -\frac{1}{y^2} \frac{dy}{dt}$$

③ Start building an equation for z:

↳ In equation ② $\frac{dy}{dt}$ is on the R.H.S

↳ $\frac{dy}{dt}$ is the original non-linear, 1st order ODE.

$$\text{i.e. } \left[\frac{dy}{dt} = ay - by^2 \right]$$

$$\frac{dz}{dt} = \left(-\frac{1}{y^2} \right) [ay - by^2] \quad \text{into ③}$$

Aside:

Harvesting problem:

↳ Other type of "special" non-linear, 1st order ODE.

$$\frac{dy}{dt} = ay - by^2 - c$$

Constants harvesting term

Harvesting term:

"Things" removed per unit time

$$\frac{dz}{dt} = \frac{-ay}{yz} + \frac{byz}{yz}$$

$$\Rightarrow \frac{dz}{dt} = -\frac{a}{y} + b$$

Recall: $z = \frac{1}{y}$; Equation ①

$$\frac{dz}{dt} = -az + b$$

Because we chose $z = \frac{1}{y}$
and followed through,
we now have a linear, 1st order
ODE

④ Solve Equation ④:

Particular solution (z_p):

$z_p = A \rightarrow$ Substitution Eq ④

$$\Rightarrow \frac{dA}{dt} = -aA + b$$

$$\Rightarrow A = \frac{b}{a} \Rightarrow \boxed{z_p = \frac{b}{a}}$$

Homogenous solution (z_h):

From equation ④

$$\frac{dz_h}{dt} = -az_h$$

$$\int \frac{dz_h}{z_h} = \int -a dt + C$$

$$\ln(z_h) = -Cat$$

$$\Rightarrow \boxed{z_h = Ce^{-at}}$$

Complete solution: $z(t) = z_p + z_h$

$$\boxed{z(t) = \frac{b}{a} + Ce^{-at}}$$

$\rightarrow$ ⑤

⑤ Convert $z(t)$ back onto $y(t)$: [Recall eq ①]

$$z(t) = \frac{aCe^{-at} + b}{a} \Rightarrow y(t) = \frac{a}{aCe^{-at} + b}$$

$$\left[z = \frac{1}{y} \right]$$

$$y(t) = \frac{a}{de^{-at} + b}$$

- $d = aC$; new constant - that's all! 
- "d" is calculated using the initial condition i.e. $y(0) = \text{something}$
- "a, b" are constants from the non-linear, 1st order, ODE

⑥ Understand and learn from the result, Eq. (6)

With $y(t) = \frac{a}{de^{-at} + b}$, what happens when $t \rightarrow \infty$ & $t \rightarrow -\infty$?

Case (i): $t \rightarrow \infty$

- " de^{-at} " becomes smaller & smaller
- " de^{-at} " becomes insignificant

$$\therefore y(\infty) = \frac{a}{de^{-at} + b} \Rightarrow \boxed{y(\infty) = \frac{a}{b}} \quad \text{or } \boxed{y = \frac{a}{b}}$$

Case (ii): $t \rightarrow -\infty$

- " de^{-at} " becomes larger and larger
- " $de^{-at} + b$ " becomes larger and larger
- $\frac{a}{de^{-at} + b}$ becomes smaller and smaller, until it becomes insignificant

$$\therefore \boxed{y(-\infty) = 0} \quad \text{or } \boxed{y = 0}$$

$$y(-\infty) = 0 \quad \& \quad y(\infty) = \frac{a}{b}$$

Represent the
Lower limit & Upper limit

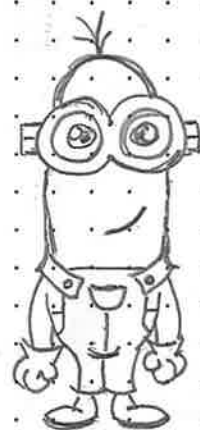
of the solution

$$y(t) = \frac{a}{de^{-at} + b}$$

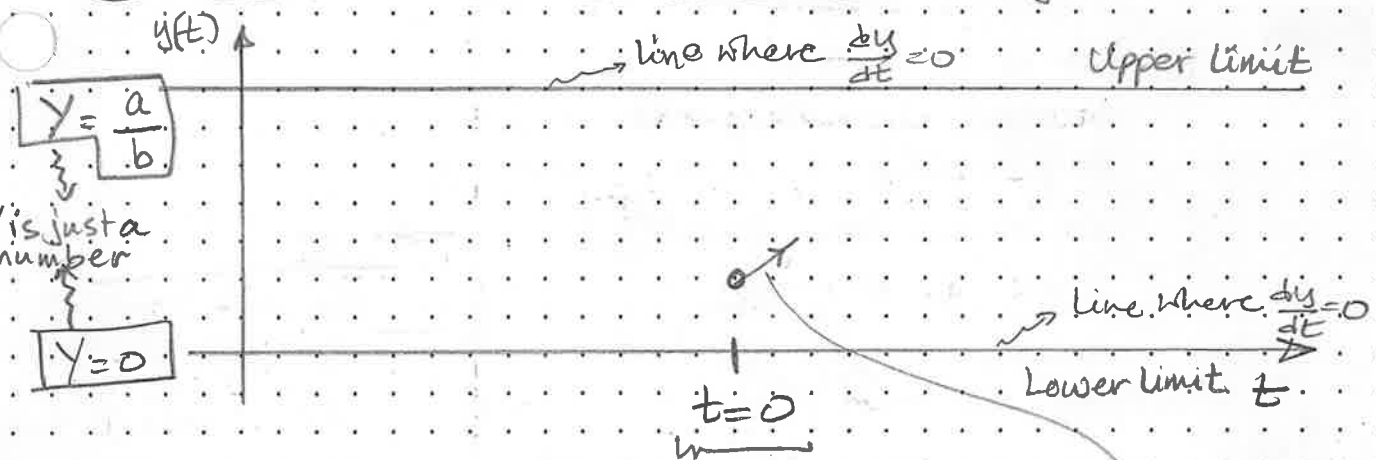
to

the non-linear, 1st order ODE

$$\frac{dy}{dt} = ay - by^2 \quad \text{Population Growth Equation}$$



(7) Construct the solution graphically:

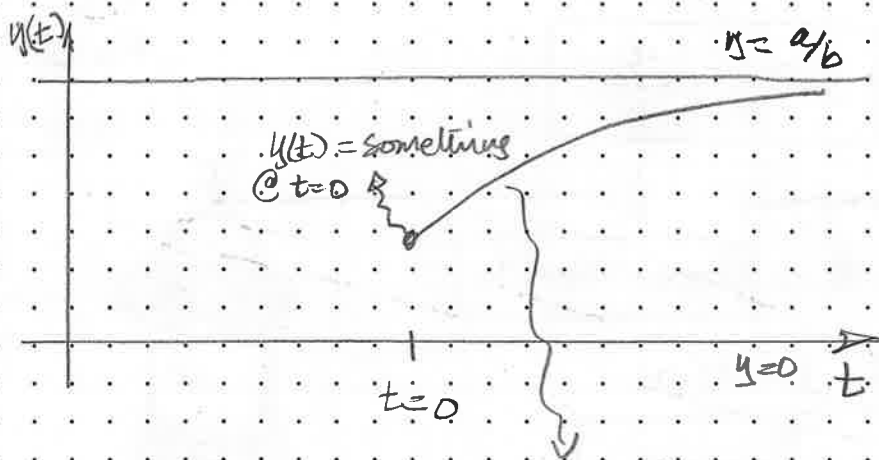


↳ We start at some time $t=0$

↳ We watch the population grow

↳ But we know that the "competition term", $(-by^2)$ will slow it down.

$\frac{dy}{dt}$ can only increase if $(ay - by^2)$ is positive.
That increase can only happen under the line $y = \frac{a}{b}$ at best.
As time progresses, $\frac{dy}{dt}$ cannot grow because by^2 will get bigger than ay i.e., $(ay - by^2) < 0$.

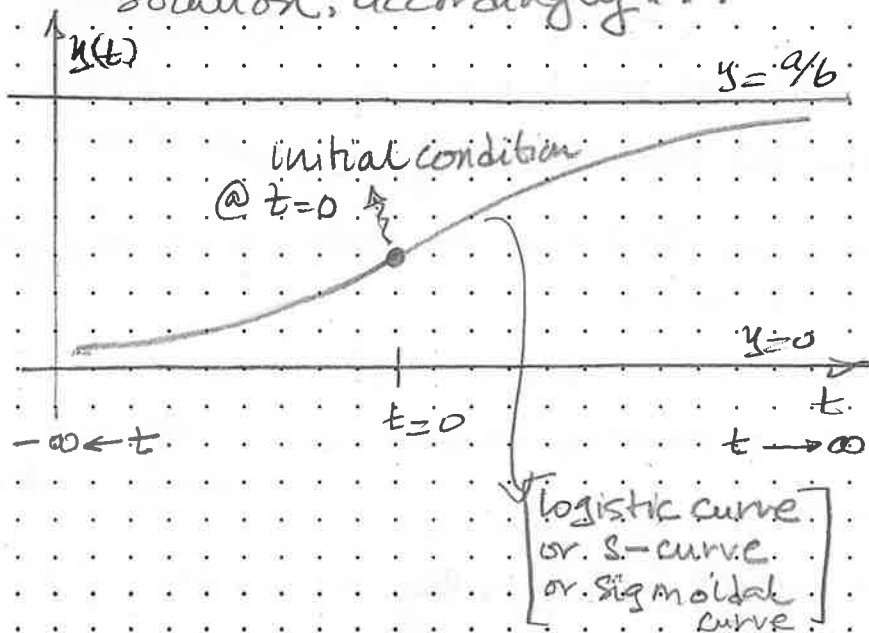


"y" can grow as long as it is under the $y = \frac{a}{b}$ line

Note:

1. $y(t)$ @ $t=0$ is "some" initial condition.
2. " $t=0$ ", can in reality be year 2000 or your birth year or whatever.
3. But if the population started actually at some time, $t = -\infty$, we know from point (6) that $y(-\infty) \rightarrow 0$.

Let's update our graphical solution, accordingly...



Aside from the main point:

At $y = \frac{a}{b}$

$$\begin{aligned} \frac{dy}{dt} &= ay - by^2 \\ &= a \frac{a}{b} - b \frac{a^2}{b^2} \\ &= \frac{a^2}{b} - \frac{a^2}{b} \end{aligned}$$

$\Rightarrow \frac{dy}{dt} = 0$

there, slope is zero at $y = \frac{a}{b}$.

At $y = \frac{2a}{b}$

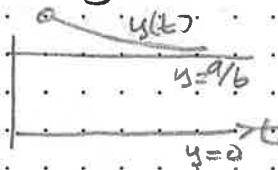
$$\begin{aligned} \frac{dy}{dt} &= a \left(\frac{2a}{b} \right) - b \left(\frac{4a^2}{b^2} \right) \\ &= \frac{2a^2}{b} - \frac{4a^2}{b} \\ &= -\frac{2a^2}{b} \end{aligned}$$

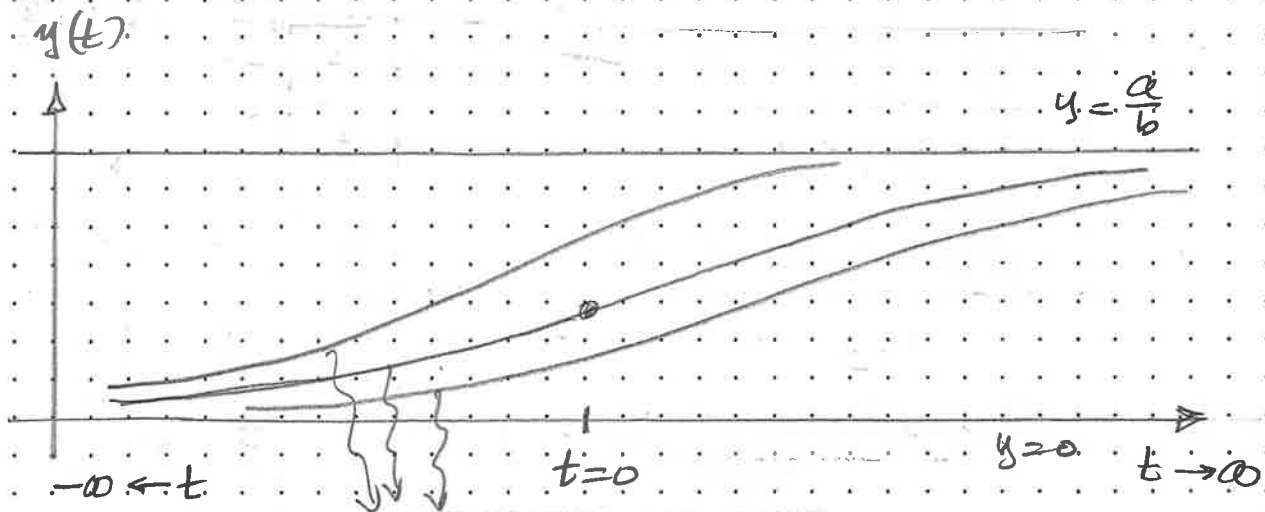
$\Rightarrow \frac{dy}{dt} < 0$

If you start at a value $y > \frac{a}{b}$.

The slope will turn negative and drive $y(t)$ toward the line $y = \frac{a}{b}$.

$y = \frac{a}{b}$





↳ S-curves depending on initial condition
 ↳ Integral lines

$$\text{STEADY STATES OF } \frac{dy}{dt} = ay - by^2$$

(Defn) $\rightarrow$ Values of $y(t)$ where $\frac{dy}{dt} = 0$

Simple explanation:

$$\frac{dy}{dt} = ay - by^2$$

$\leftarrow$ This is our non-linear, 1st order ODE

⊕ We want to evaluate it at $\frac{dy}{dt} = 0$ for our steady states

⊕ Say we have a number, y @ $\frac{dy}{dt} = 0$

$$y: ay - by^2 = 0$$

$$\Rightarrow y(a - by) = 0$$

$$\Rightarrow y = 0$$

$$\text{or } y = \frac{a}{b}$$

Steady state ($y = 0$)

$\rightarrow$ If we stay at 0, nothing will happen.

$\rightarrow$ No population or

$\rightarrow$ No population growth

Steady State ($y = \frac{a}{b}$)

$\rightarrow$ If we start at $y = \frac{a}{b}$, we will stay at $\frac{a}{b}$

If we move a little away from $y = 0$

$\rightarrow$ we grow exponentially

$\rightarrow$ $-by^2$ term kicks in

$\rightarrow$ we slow down the growth

$\rightarrow$ And reach $\frac{a}{b}$

Final take home message about steady states.

$$\frac{dy}{dt} = ay - by^2$$

$$y=0$$

→ If we are a little above $y=0$, we "take off" and move away from $y=0$

Unstable steady state

$$y = \frac{a}{b}$$

→ If we are near $y = a/b$, we get closer to a/b

Stable steady state

STABILITY/INSTABILITY OF STEADY STATES

→ There is more systematic way of assessing stability of the steady-states!

→ Say we have a non-linear, 1st order ODE

$$\frac{dy}{dt} = f(y)$$

→ Find Steady State: $\frac{dy}{dt} = 0 \Rightarrow f(y) = 0$

→ Find y such that $f(y) = 0$

→ To evaluate stability of steady states

→ Look for $\frac{df}{dy}$ at $y = y^*$

$$\text{If } \left. \frac{df}{dy} \right|_{y=y^*} < 0 \rightarrow \text{stable}$$

EXAMPLES

$$\frac{dy}{dt} = ay$$

$$\Rightarrow y(t) = e^{at}$$

$$ay = 0$$

$$\Rightarrow y = 0$$

One steady state

Assess steady states (1)

$$f = ay$$

$$\frac{df}{dy} = a$$

$$\boxed{\text{Stable if } a < 0}$$

Stable if

$$\left(\frac{df}{dy} < 0 \right)$$

$$\frac{dy}{dt} = y - y^2$$

$$y - y^2 = 0$$

$$\Rightarrow y(1 - y) = 0$$

$$\Rightarrow y = 0 \quad y = 1$$

Two steady states

$$f = y - y^2$$

$$\frac{df}{dy} = 1 - 2y$$

$$\text{At } y = 0:$$

$$\left[\frac{df}{dy} = 1 > 0 \right] \text{ Unstable}$$

$$\text{At } y = 1:$$

$$\left[\frac{df}{dy} = -1 < 0 \right] \text{ Stable}$$

$$\frac{dy}{dt} = y - y^3$$

$$y - y^3 = 0$$

$$\Rightarrow y(1 - y^2) = 0$$

$$\Rightarrow y = 0 \quad y = 1 \quad y = -1$$

Three steady states

$$f = y - y^3$$

$$\frac{df}{dy} = 1 - 3y^2$$

$$\left[\text{At } y = 0: \frac{df}{dy} = 1 > 0 \right] \text{ Unstable}$$

$$\left[\text{At } y = 1: \frac{df}{dy} = -2 < 0 \right] \text{ Stable}$$

$$\left[\text{At } y = -1: \frac{df}{dy} = -2 < 0 \right] \text{ Stable}$$

65

KvB